

Aditya K

 XGK-Loa8Driller |  aditya-k |  aditya.k31206@gmail.com |  +91 9566083914

SUMMARY

ECE student with an interest in embedded systems, programming, and digital electronics. Skilled in C, C++, Verilog, and hardware-software integration, with hands-on experience through projects and an Embedded Systems internship at Data Patterns. Passionate about learning new technologies and building practical engineering solutions.

EDUCATION

2024 - present	B.E. (ECE) at St. Joseph's College of Engineering	(CGPA: 7.97)
2024	Class 12th CBSE Board	(81.8%)
2022	Class 10th CBSE Board	(89.8%)

INTERNSHIP EXPERIENCE

Data Patterns – Embedded Systems Intern

- Gained hands-on exposure to real-world defence and aerospace electronics development.
- Worked with embedded hardware, microcontrollers, sensors, and communication interfaces for hardware-software integration.
- Acquired practical experience in real-time embedded systems, system testing, troubleshooting, and electronic system design.
- Strengthened ability to translate engineering concepts into reliable, real-world solutions.

IIITDM Kurnool – Ball Drone MkII Project

- Designed and developed a spherical UAV with a focus on omnidirectional mobility and agile flight control systems.
- Integrated Radiolink F722 flight controller, brushless motor, ESC, servos, and Li-Po battery.
- Contributed to flight controller configuration and overall hardware integration.
- Gained strong understanding of aerodynamics, servo-based actuation, motor control, and real-time flight stabilization techniques.

PROJECTS

Raksha Tejas – Anti-Drone Sentinel Rover System

Engineered a cutting-edge anti-drone defense concept designed to detect, track, and respond to unauthorized UAV activity in real time using a fusion of advanced sensors and embedded systems. Focused on seamless hardware-software integration to enable rapid threat identification and situational awareness, while exploring RF signal detection, secure communication protocols, and intelligent automated response mechanisms for robust aerial security.

Ball Drone MkII – Omni-Directional Lightweight Drone

Developed a spherical UAV engineered for extreme agility and true omnidirectional movement. Its unique enclosed spherical architecture enables seamless 360° motion—forward, backward, lateral, and rotational—allowing it to maneuver through complex environments with fluid precision. The design delivers highly responsive, collision-resistant flight behavior and unmatched maneuverability in all directions.

A.R.T.E.M.I.S. – Railway Track Monitoring System

Developed a high-impact AI railway inspection system that fuses computer vision and vibration analytics to instantly detect critical track failures like cracks, missing fasteners, fishplate defects, and geometric distortions. Leveraged YOLO-based real-time object detection with a camera system and an IMU sensor for motion/vibration sensing, all processed on the NVIDIA Jetson Orin Nano for edge-level intelligence. Enables autonomous, high-speed track surveillance with early fault prediction.

CERTIFICATIONS

- NPTEL** Introduction to Programming in C
- NPTEL** Programming in Java
- NPTEL** Problem Solving through Programming in C
- Oracle** Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate

TECHNICAL SKILLS

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|-------------------|--|
| Languages | C, C++, Verilog, Java |
| Embedded Systems | Microcontrollers, Sensors, Communication Interfaces, Hardware-Software Integration |
| Tools & Platforms | NVIDIA Jetson, YOLO, Flight Controllers (Radiolink F722), ESC, Servos |
| Domains | Digital Electronics, Computer Vision, RF Detection, Real-Time Systems, UAV Design |

SOFT SKILLS

Teamwork & Collaboration | Leadership | Problem Solving | Communication | Adaptability